
Problem 8.27. Recall that a directed graph is strongly connected if every two nodes are connected by a directed path in each direction. Let

$$STRONGLY\text{-}CONNECTED = \{\langle G \rangle \mid G \text{ is a strongly connected graph}\}.$$

Show that $STRONGLY\text{-}CONNECTED$ is NL-complete.

Proof. To show that $STRONGLY\text{-}CONNECTED$ is NL-complete, we must show that it is in NL and that all NL problems are log space reducible to it. To show $STRONGLY\text{-}CONNECTED \in$ NL, we give an algorithm S that decides $\overline{STRONGLY\text{-}CONNECTED}$ in non-deterministic log space.

$S =$ "On input $\langle G \rangle$, where G is a directed graph:

1. Non-deterministically select every distinct pair of nodes s and t in G:
2. Run the algorithm M for $\overline{PATH}$ given in Theorem 8.27 on $\langle G, s, t \rangle$.
3. *Accept*, if M accepts. *Reject*, if M rejects."

To prove that $STRONGLY\text{-}CONNECTED$ is NL-hard, we give a log space reduction from $PATH$ to $STRONGLY\text{-}CONNECTED$. This reduction converts a directed graph G and two nodes s and t to a directed graph G' so that G has directed path from s to t, iff G' is strongly connected. Next, we show how to construct the graph G'. The graph G' is same as the graph G, but includes following additional edges, which we name connectivity edges:

1. An edge to s from every node in G except t.
2. An edge from t to every node in G except s.

Suppose graph G contains a path from node s to t. Then, G' is strongly connected, because a path from any two distinct nodes u and v in G' can be constructed as follows:

1. Start at node u, and go to s.
2. Follow the path from s to t.
3. Go to node v.

Suppose G' is strongly connected. Then, there are two distinct nodes s and t in G', such that there is an edge to s from every other node in G' except t, and there is an edge from t to every other node in G' except s. G' also contains a path from node s to t, and removing connectivity edges from G' to construct G does not remove this path from s to t in G. A log space transducer can easily implement construction of G' from G, s and t. Hence, $STRONGLY\text{-}CONNECTED$ is NL-hard. □